Comparative Analysis Of Machine Learning Classifiers for Phishing URL Detection:A Security-Oriented Evaluation

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ABSTRACT

Phishing URL detection is a critical component of modern web security infrastructure.
Machine Learning solutions are capable of detecting attack patterns independent of prior domain knowledge , surpassing signature approaches that fail against zero-day phishing.
This work presents a comparative evaluation of four classification algorithms: Logistic Regression,Naive Bayes,Random Forest, and XGBoost on a dataset of **11,055** labeled URLs across **30** security-engineered features. We evaluate models using accuracy , F1-score,ROC-AUC, and precision recall tradeoff with consideration towards cost of false negatives in security context.Ensemble methods achieve **97%** accuracy and **0.996** ROC-AUC , outperforming linear models.Naive Bayes , despite competitive ROC-AUC of **0.966** , exhibits critical failure in deployment relevant metric due to violated feature independence assumptions,We analyse these results through a security operations lens,identifying deployment appropriate threshold for a browser extension context.

INTRODUCTION

Phishing URL detection has become a critical research problem as signature-based blacklist approaches fail against zero-day domains i.e; newly registered phishing sites absent from known threat databases. Intruders continuously evolve their techniques across networks, psychology, and architecture, meaning we can never fully anticipate the unknown. Machine learning offers a dynamic alternative: by learning structural patterns across URL features, models can detect unseen phishing attempts without prior domain knowledge [1].
The core challenge is asymmetric cost. A false negative (a missed phishing URL) carries fundamentally higher operational cost than a false positive. Researchers have traditionally evaluated phishing detection models on accuracy or F1-score alone, but a single metric never correlates reliably to successful deployment. Every model has an inherent tradeoff between its False Positive Rate (FPR) and False Negative Rate (FNR), and that tradeoff must be explicitly analysed [2].

This study compares four classifiers; Logistic Regression, Naive Bayes, Random Forest, and XGBoost , evaluating tradeoffs between linear and ensemble approaches with respect to threshold behaviour, feature correlation, and security-oriented deployment constraints.

APPROACH

The dataset '***Phishing URL detection dataset***' [3] aggregates **11,055** labeled URLs from phishing reports, threat intelligence platforms, and legitimate URL repositories. Features are pre-engineered and encoded as -1 (phishing indicator), 0 (uncertain), or 1 (legitimate indicator) across 30 signals spanning URL structure, SSL state, domain registration, and external reputation. Chi-squared and mutual information analysis identified SSLfinal_State, URL_of_Anchor, Prefix_Suffix, and web_traffic as top discriminative features as shown in the feature importance analysis graph (*see Figure A3 in Appendix A*). All **30** features were retained for fair cross-algorithm comparison. The dataset was partitioned **80/20** (**8,844** training, **2,211** test) with stratification and random_state=42.

Confusion Matrix

The pre-engineered features are binary classified , the models involve deals with the two classes i.e, 1 and -1 for legitimate and phishing respectively while 0 suggest uncertainty.The confusion matrix represented in this study is of 2x2 (*see Table A1 in Appendix A*)

METHODS

Logistic Regression

Logistic Regression models the probability of class membership by fitting a linear decision boundary in feature space using maximum likelihood estimation. Coefficients are optimised iteratively via gradient descent until convergence, making max_iter a critical hyperparameter. As a linear model, it assumes phishing and legitimate URLs are separable by a hyperplane across 30 features, an assumption we empirically evaluate. Its primary advantage is interpretability: each coefficient directly quantifies a feature's contribution to the log-odds of phishing classification.

Naive Bayes

Naive Bayes applies Bayes' theorem to compute posterior class probabilities, assuming conditional independence between all features given the class label. This assumption makes training computationally trivial , parameters are estimated directly from feature distributions in a single pass

with no iteration. However, URL features violate this assumption structurally: SSLfinal_State, age_of_domain, and Page_Rank are correlated signals that co-occur in both phishing and legitimate URLs. When correlated evidence is treated as independent, Naive Bayes multiplies overlapping signals, distorting probability estimates and producing miscalibrated predictions.

Random Forest

Random Forest constructs an ensemble of decision trees, each trained on a random bootstrap sample of training data and a random subset of features at each split node. Classification is determined by majority vote across all trees. This architecture provides two key advantages: correlated errors across trees are reduced by feature randomisation, and weak features are naturally deprioritised when excluded from informative splits. Random Forest performs feature selection during training.

XGBoost

XGBoost implements gradient boosted decision trees, constructing an ensemble sequentially where each tree corrects residual errors of the preceding ensemble [4]. Unlike Random Forest's parallel bagging, XGBoost uses second-order Taylor expansion of the loss function to compute precise gradient and curvature information, enabling more efficient tree construction. L1 and L2 regularisation terms penalise model complexity, reducing overfitting on noisy features. This combination of sequential error correction and regularisation makes XGBoost particularly effective on structured tabular data with mixed feature relevance.

RESULTS

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	0.92	0.92	0.90	0.91	0.977
NB	0.58	0.51	1.00	0.67	0.966
RF	0.97	0.97	0.95	0.96	0.996
XGB	0.97	0.98	0.95	0.97	0.996

Ensemble methods substantially outperform linear classifiers, with Random Forest and XGBoost achieving **97%** accuracy and **0.996** ROC-AUC (*Figure A2 in Appendix A*) compared to Logistic Regression's **92%** and **0.977.** This performance gap is consistent with a nonlinear decision boundary in URL feature space; phishing detection cannot be reduced to a single separating hyperplane across 30 correlated features.

Naive Bayes presents the most analytically significant result. Despite achieving **0.966** ROC-AUC competitive with ensemble methods it produces only **58%** accuracy with a **72%** false positive rate on legitimate URLs. Its confusion matrix reveals **921** legitimate URLs misclassified as phishing out of **1,255.** This paradox arises because ROC-AUC measures probability ranking quality across all thresholds, while accuracy reflects performance at the default **0.5** threshold. Naive Bayes ranks URLs reasonably but its probability estimates are severely miscalibrated due to violated independence assumptions, pushing predictions toward the phishing class regardless of threshold.

From a deployment perspective, we analyse operating threshold selection for a browser extension context. At **2-3%** false positive rate, XGBoost achieves greater than **95%** phishing recall , catching **19** out of every **20** phishing URLs while wrongly flagging approximately one legitimate site per **50** visited. This asymmetric operating point is appropriate given that the consequence of a missed phishing attack (credential theft, financial loss) substantially exceeds the cost of a false alarm (user friction). Logistic Regression requires a higher false positive rate to achieve equivalent recall, limiting its suitability for this deployment despite its interpretability advantage.

CONCLUSION

This study evaluated four classifiers for phishing URL detection across accuracy, F1-score, ROC-AUC, and security-relevant deployment metrics. Ensemble methods ; Random Forest and XGBoost achieved superior performance (**97%** accuracy, **0.996** ROC-AUC), outperforming linear models due to the nonlinear structure of the phishing detection boundary. Naive Bayes demonstrated that ROC-AUC alone is insufficient for evaluating security classifiers: high ranking quality can coexist with catastrophic threshold miscalibration when feature independence assumptions are violated.

Future work will first explore feature engineering from raw URL strings to reduce dependence on fixed pipelines. Second, adversarial robustness will be tested against evasion attacks. Ultimately, this research aims to bridge ML security and threat modelling by investigating architectures with unique computational properties, such as Spiking Neural Networks.

REFERENCES

[1] O. K. Sahingoz, E. Buber, O. Demir, and B. Diri, "Machine learning based phishing detection from URLs," Expert Systems with Applications, 2019 (p. 3).

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.**[3]** H. Ullah, "Phishing URL detection dataset," Kaggle, 2024. (Online). Available: (URL) (p. 3).

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," Proceedings of KDD, 2016 (p. 3).

Appendix A:

Model Abbreviations & Evaluation Framework

The following abbreviations are used throughout this study to represent the evaluated machine learning architectures:

LR: Logistic Regression (Linear Baseline)
NB: Naive Bayes (Probabilistic Classifier)
RF: Random Forest (Ensemble Bagging)
XGB: XGBoost (Extreme Gradient Boosting)

Hyperparameters:

LR: max_iter=**1000**. (p. 1)
NB: Default GaussianNB. (p. 1)
RF: n_estimators=**100**, random_state=**42**. (p. 2)
XGB: learning_rate=**0.1**, n_estimators=**100**. (p. 2)

Table A1: ***Confusion Matrix (p .1)***

The confusion matrix below defines the classification outcomes used to calculate security-relevant metrics.

		Predicted Values	
		Phishing (-1)	Legitimate (1)
Actual Values	Phishing (-1)	True positive	False Positive
	Legitimate (1)	False Negative	True Negative

FigureA2: ***Comparison of ROC curves across all four classifiers*** **(p .2)**

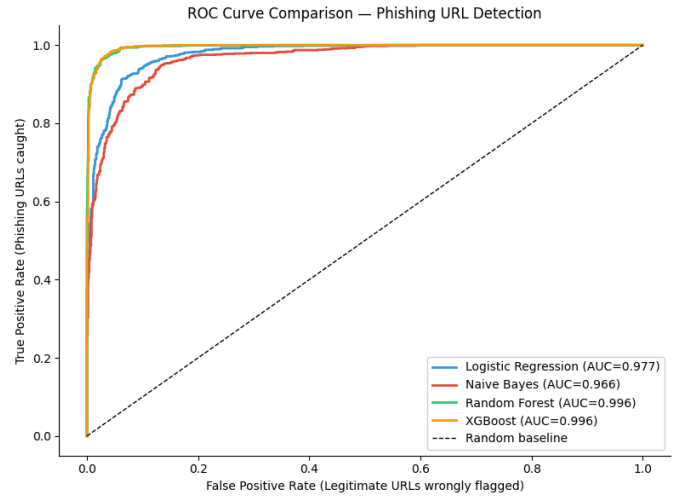
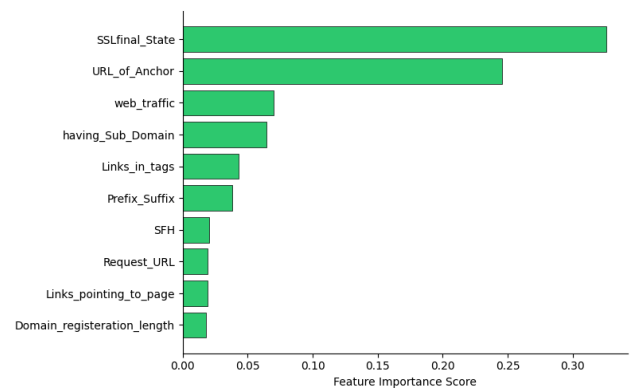


Figure A3: ***Feature Importance Analysis (p .1)***

The model utilized 30 pre-encoded features including SSLfinal_State, URL_of_Anchor, and Prefix_Suffix